Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

Metric	Value
Proteins parsed	6
Molecular-weight bins	4
Bin width	2 kDa
MW range	36.10 to 58.01 kDa
Median MW	41.57 kDa
Mean MW	45.68 kDa
Proteins <= 50 kDa	4 (66.7%)
Proteins 50-100 kDa	2 (33.3%)
Proteins > 100 kDa	0 (0.0%)
High-priority bins	1
Elevated-priority bins	0
Routine-priority bins	3
Source FASTA	demo_human_subset.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

Key plot

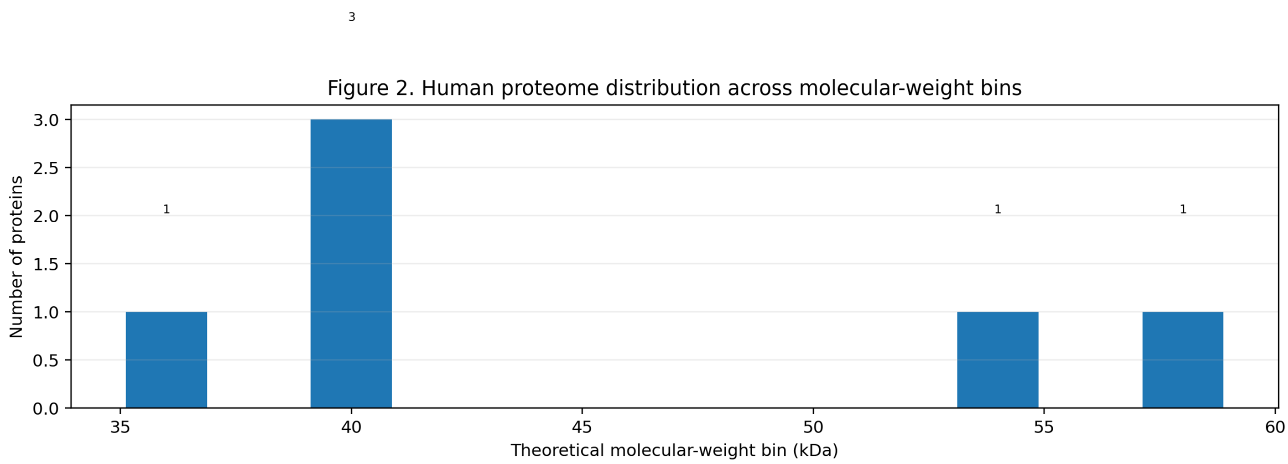


Figure 2. Human proteome distribution across molecular-weight bins

Figure 2. Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.

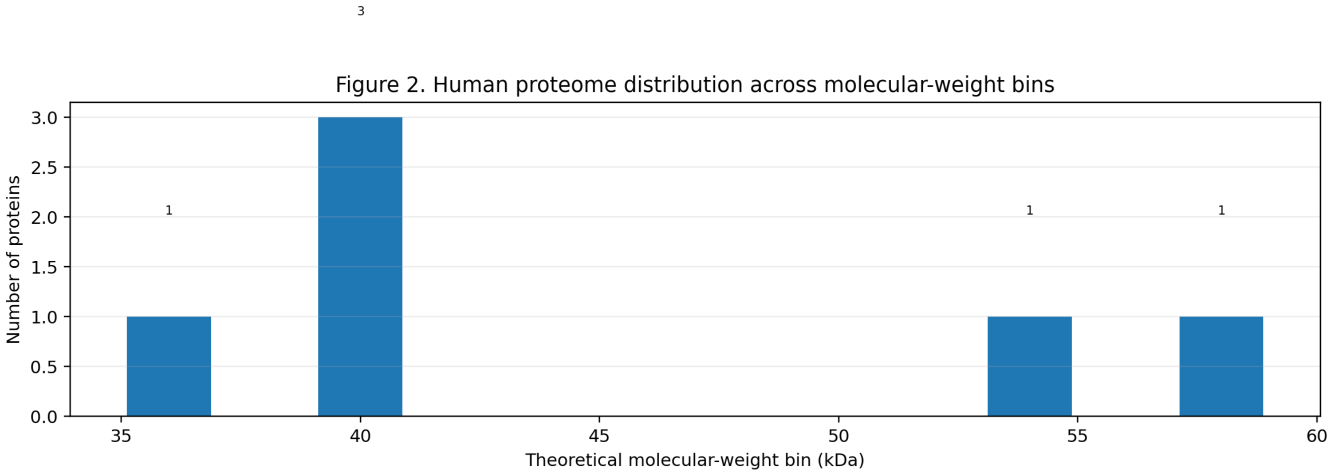


Figure 3. Most crowded molecular-weight neighborhoods.

The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.

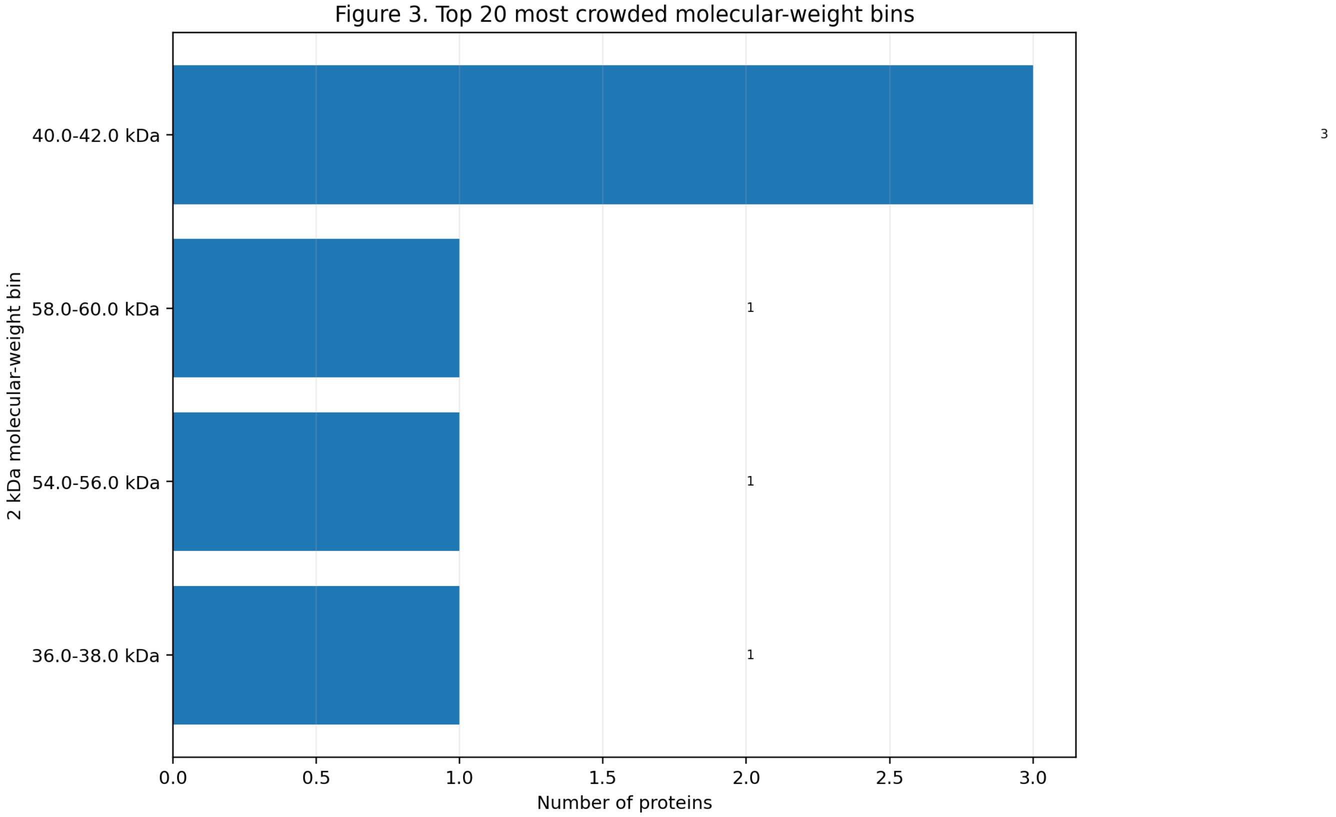


Figure 4. Distribution of common western blot markers and signaling proteins across molecular-weight bins.

The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

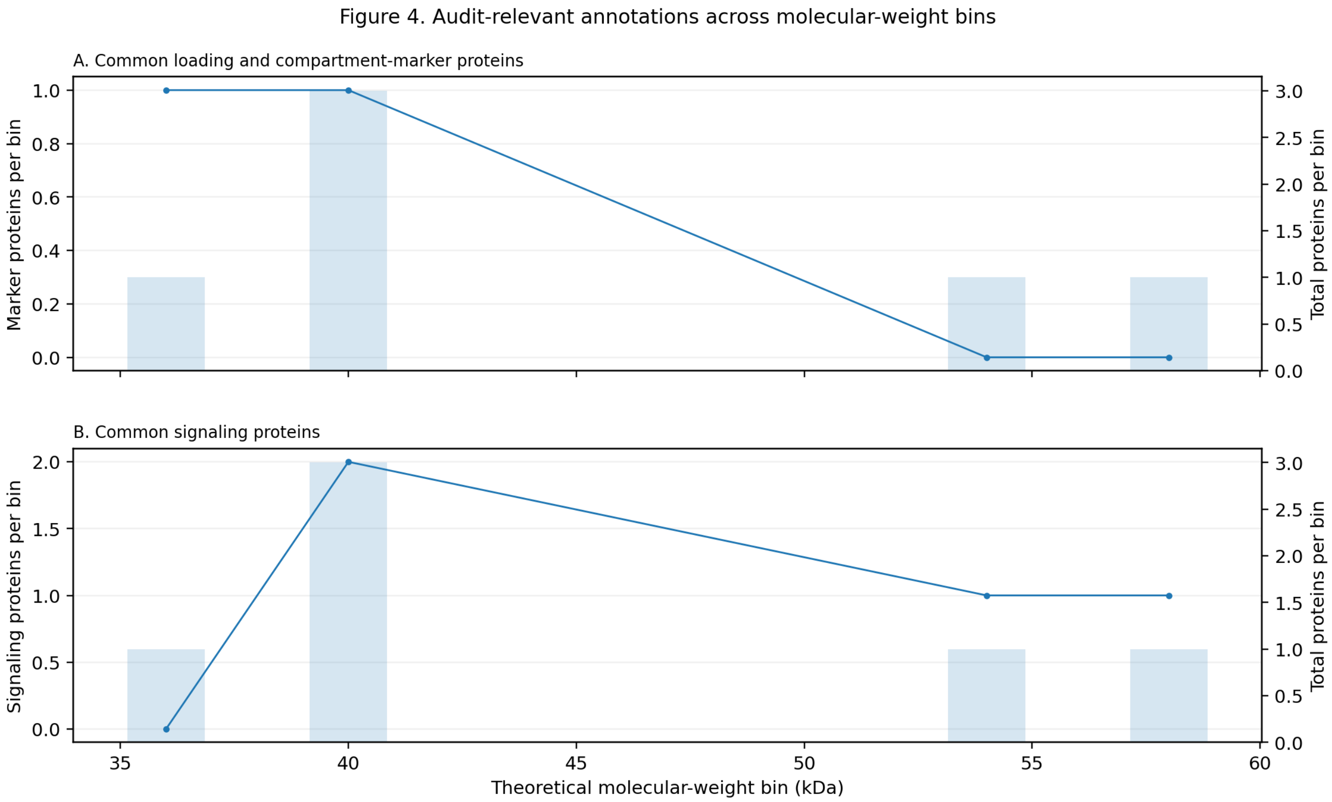
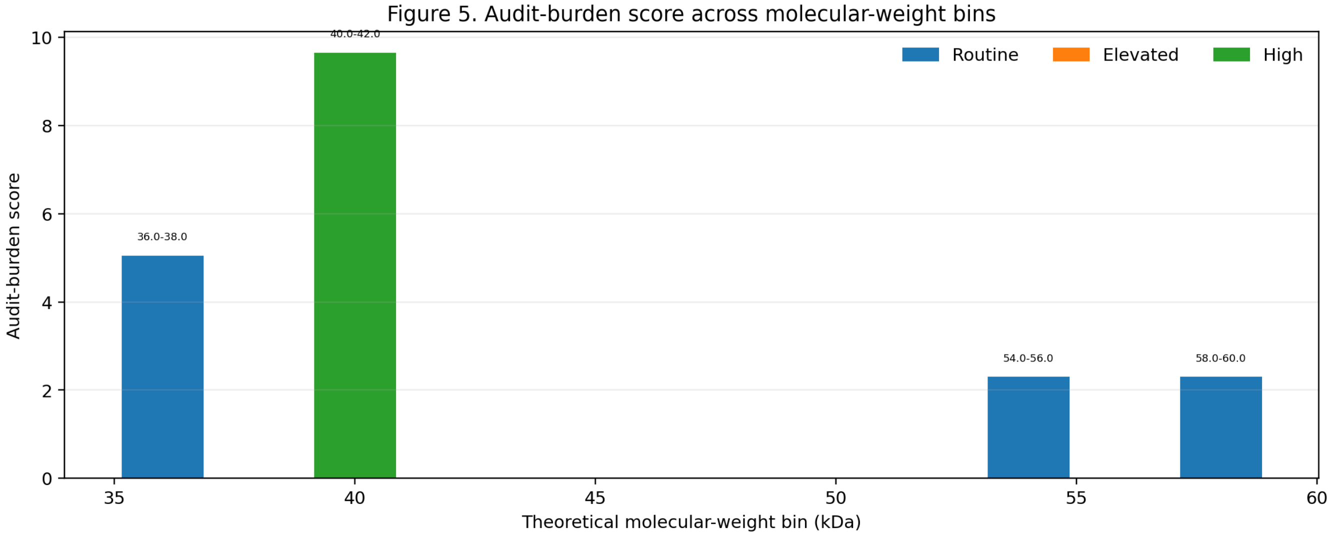


Figure 5. Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
40.0-42.0 kDa	3	1	2	high	9.7
36.0-38.0 kDa	1	1	0	routine	5.0
54.0-56.0 kDa	1	0	1	routine	2.3
58.0-60.0 kDa	1	0	1	routine	2.3

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
40.0-42.0 kDa	3	1	2	Actin family; MAPK cascade	high	MAPK14 [MAPK cascade], MAPK1 [MAPK cascade], ACTB [Actin family]
36.0-38.0 kDa	1	1	0		routine	GAPDH
54.0-56.0 kDa	1	0	1	AKT pathway	routine	AKT1 [AKT pathway]
58.0-60.0 kDa	1	0	1	JAK-STAT	routine	STAT1 [JAK-STAT]

Common loading/compartment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
36.10	GAPDH	P04406	G3P_HUMAN	Glyceraldehyde-3-phosphate dehydrogenase	36.0-38.0 kDa
41.78	ACTB	P60709	ACTB_HUMAN	Actin, cytoplasmic 1	40.0-42.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
36.0-38.0 kDa	1	1	0		routine	GAPDH
40.0-42.0 kDa	3	1	2	Actin family; MAPK cascade	high	MAPK14 [MAPK cascade], MAPK1 [MAPK cascade], ACTB [Actin family]
54.0-56.0 kDa	1	0	1	AKT pathway	routine	AKT1 [AKT pathway]
58.0-60.0 kDa	1	0	1	JAK-STAT	routine	STAT1 [JAK-STAT]

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
36.10	GAPDH	P04406	Glyceraldehyde-3-phosphate dehydrogenase	36.0-38.0 kDa		configured marker/control; verify source-file provenance if used near target bands
41.29	MAPK14	Q16539	Mitogen-activated protein kinase 14	40.0-42.0 kDa	MAPK cascade	configured signaling gene; check phospho/total controls and antibody validation; configured audit family match: MAPK cascade; crowded MW neighborhood; require full blot, ladder, raw metadata, and controls
41.36	MAPK1	P28482	Mitogen-activated protein kinase 1	40.0-42.0 kDa	MAPK cascade	configured signaling gene; check phospho/total controls and antibody validation; configured audit family match: MAPK cascade; crowded MW neighborhood; require full blot, ladder, raw metadata, and controls
41.78	ACTB	P60709	Actin, cytoplasmic 1	40.0-42.0 kDa	Actin family	configured marker/control; verify source-file provenance if used near target bands; configured audit family match: Actin family; crowded MW neighborhood; require full blot, ladder, raw metadata, and controls
55.54	AKT1	P31749	RAC-alpha serine/threonine-protein kinase	54.0-56.0 kDa	AKT pathway	configured signaling gene; check phospho/total controls and antibody validation; configured audit family match: AKT pathway
58.01	STAT1	P42224	Signal transducer and activator of transcription 1-alpha/beta	58.0-60.0 kDa	JAK-STAT	configured signaling gene; check phospho/total controls and antibody validation; configured audit family match: JAK-STAT

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38δ western blot	40.0-42.0 kDa	ORI reported p38δ bands were compiled from a source image representing β-actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38δ-labeled bands were compiled from a β-actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_GLUT4	Nrf2, NF-kB p65, PPARα/δ, Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	demo_human_subset.fasta
Marker file	human_marker_genes.tsv
Signaling-prefix file	human_signaling_prefixes.tsv
Family-pattern file	human_family_patterns.tsv
Species/report label	Homo sapiens demo subset